

Secure Tracking of Patient's Vital Signs Using CSI-Based Homomorphic Encryption-Enabled Deep Learning Framework

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Abstract—Preserving patient privacy in digital healthcare systems is a critical challenge, particularly in non-intrusive monitoring applications. This paper introduces VitalCrypt, a novel framework for secure and real-time vital sign monitoring that combines Channel State Information (CSI) with homomorphic encryption and lightweight deep learning. Homomorphic encryption enables computations directly on encrypted data, ensuring data confidentiality throughout the processing pipeline. The framework incorporates well-established signal preprocessing techniques, such as Hampel, Savitzky-Golay, and elliptic filters for noise removal, with Principal Component Analysis (PCA) for dimensionality reduction. The Power Spectral Density (PSD) of these refined signals is used as features, which are then fed into a lightweight neural network optimized with encryption-compatible activation functions for classification. The system effectively classifies breathing and heart rates while maintaining compatibility with homomorphic encryption schemes. Experimental evaluations were conducted using a publicly available dataset. The results demonstrated exceptional accuracy, achieving 99.46% for breathing rate classification on plain data and 99.44% on encrypted data, with negligible performance degradation despite increased runtime due to encryption. The results of heart rate classification are also discussed. The framework processes encrypted data at approximately seven times the latency of plain data; however, this trade-off is justified by the substantial privacy benefits attained. VitalCrypt showcases the potential of secure, privacy-preserving deep learning applications in healthcare, addressing critical challenges in real-time, non-intrusive patient monitoring. By balancing high accuracy and data confidentiality, this framework provides a scalable solution for healthcare applications, including remote monitoring and clinical diagnostics.

Index Terms—CSI, lightweight neural network, homomorphic encryption, privacy-preserving, vital sign classification.

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I. INTRODUCTION

NOWADAYS, non-intrusive systems are getting really popular due to their observation capability with minimal physical interaction or disruption to the subject, the environment, or the process that is being observed. Due to its high potential impact, researchers are constantly focusing on designing non-intrusive systems for different purposes, such as driver drowsiness monitoring [1], sleep staging [2], vital sign monitoring [3], and many more. Non-intrusive vital sign monitoring systems have emerged as transformative solutions in healthcare, leveraging technologies such as cameras [4], radar [4], and infrared sensors [5] to measure essential health parameters without direct physical contact. These systems enhance patient comfort by minimizing invasiveness, reducing the risk of cross-contamination, and enabling continuous, real-time monitoring. Particularly advantageous in sensitive environments such as neonatal care or remote healthcare settings, non-intrusive monitoring facilitates early detection of critical conditions like cardiovascular or respiratory diseases, improves patient safety, and enhances the overall efficiency and accessibility of healthcare services [6].

In today's tech-driven world, CSI can be a powerful non-intrusive technology that has the potential to play a significant role in modern healthcare. By analyzing key parameters of the wireless signal—such as amplitude, phase, and delay of CSI can allow for indoor localization [7], [8], [9], [10], human activity recognition [11], vital sign monitoring [12], and many more. This makes CSI an ideal choice for applications like human presence detection, object tracking, and environmental monitoring, where traditional sensing methods may be invasive or cumbersome. CSI seamlessly integrates with existing Wi-Fi networks, offering a cost-effective and scalable solution that works well in complex, cluttered settings. Its ability to provide continuous, real-time data without impacting user comfort makes it a game-changer in sectors like smart homes, healthcare, and beyond. The versatility and non-intrusiveness of CSI empower better, more accessible, and safer monitoring across various use cases, paving the way for innovative advancements in everyday technology.

The applications of machine learning and deep learning algorithms are increasing exponentially day by day due to their enhanced capability of learning complex patterns from input features. As these algorithms are complex and computationally expensive, they often suffer from dimensionality. Input features

with a high number of dimensions can significantly enhance the computational costs, potentially making the system inefficient. PCA can be a handy tool to handle the dimensionality issue. The process can significantly reduce computational cost, making the Machine Learning (ML)-based or Deep Learning (DL)-based predictive systems faster and more efficient. CSI data contains multiple streams of time series as the system contains multiple sub-carriers from each channel. Research shows that PCA can be effectively used for efficient Human Activity Recognition (HAR)-based systems [13], [14]. In the case of vital sign detection from CSI, multiple streams of data need to be utilized for accurate classification. PCA can assist in reducing the dimensions without degrading the classification accuracy [15].

Modern healthcare systems are presently adopting ML and DL to improve the overall productivity of the healthcare sector. It is known to all that ML-based applications often require a large amount of data to work. In this case, patient data needs to be utilized for training the ML models or DL models and performing predictions on them. However, it is critical to protect the confidentiality of a patient's data [16]. With the exponential growth of data volume, digital healthcare is becoming more vulnerable to cyber threats and information breaches [17]. So, an encryption-based system combined with the power of machine learning or deep learning would greatly enhance the reliability of digital healthcare.

Homomorphic encryption is an effective scheme for applying encryption-based systems in order to address the privacy concerns of modern healthcare systems [18]. The homomorphic encryption system combined with deep learning will not only enhance the productivity of present healthcare but also ensure the confidentiality of patients' delicate information concerning their privacy. However, combining deep learning with homomorphic encryption is difficult because of its intense computational complexity [19]. So, the commencement of a lightweight, homomorphic-friendly deep learning model will add significant value to modern healthcare infrastructure. The lightweight model is not only suitable for a faster training process but also makes it more suitable for a homomorphic encryption-based system prediction system. Many studies have been done recently proposing homomorphic encryption-friendly deep neural networks for different purposes such as image classification [20], healthcare and genomics data classification [21], audio classification [22], etc.

In this research, VitalCrypt, a CSI-based homomorphic encryption-friendly deep learning framework, is proposed for vital sign monitoring, specifically focusing on breathing rate and heart rate classification. We first employ PCA on the obtained CSI data and acquire the power spectral density of the selected PCA's reducing the dimensionality. Then we utilize that power spectral density data to train our lightweight, homomorphic encryption-friendly neural network with customized activation functions. Furthermore, we evaluate our model on both plain data and encrypted data to ensure the robustness and effectiveness of our proposed neural network model with the proposed framework. The contributions of our work are as follows:

- A PCA-based privacy-preserving deep learning framework, VitalCrypt, is proposed for extracting effective features to classify breathing and heart rates from CSI data.

- A lightweight deep learning model, optimized for compatibility with homomorphic encryption, is developed. The model demonstrates high accuracy in classifying both breathing and heart rates while maintaining encryption scheme support.
- The performance of the proposed framework is evaluated and compared using both plaintext and encrypted data.
- A publicly available dataset is utilized to validate the effectiveness, reliability, and superior performance of the privacy-preserving deep learning model and framework.

II. LITERATURE REVIEW

The use of CSI has gained traction as an effective modality for contactless physiological monitoring. By analyzing fine-grained variations in WiFi signals, researchers have demonstrated promising results in respiration and heart rate tracking across diverse settings. Khan et al. [15] and Armenta-Garcia et al. [23] validated the viability of CSI for vital sign monitoring, while others like Wang et al. [24], Madhi et al. [25], and Sun et al. [26] explored CSI's sensitivity to subtle motion and biometrics. These studies collectively emphasize CSI's potential for real-time, non-intrusive applications, though most depend on high-frequency data, hardware complexity, or simplified signal assumptions that limit adaptability.

To address signal variability, researchers have incorporated learning-based models. PCA-based dimensionality reduction, as shown by Li et al. [27] and Tian et al. [28], is commonly used to preprocess CSI before classification. CNN-based systems such as the ones by Butyrskaya [29] and Madhi et al. [25] have explored spatial and temporal patterns, improving detection robustness. Additionally, visual-based CSI interpretations [29] offered novel representations but often introduced latency. These solutions, although technically sound, do not address the growing demand for privacy, especially in health-related deployments.

A substantial portion of CSI-based systems operates without embedded security protocols. For example, Alzaabi et al. [30] and Khan et al. [31] proposed practical CSI applications in patient care but did not integrate privacy safeguards. In response, homomorphic encryption (HE) has become an emerging solution for secure inference. [32] Liu et al. [33] presented one of the early works combining HE with DL in healthcare. More recently, Asif et al. [34], Ullah et al. [35], and Ullah et al. [36] demonstrated HE's potential across encrypted ECG, MRI, and diagnostic imaging, achieving high accuracy while preserving confidentiality.

Despite these advancements, a unified framework that integrates low-rate CSI processing, deep learning, and encrypted inference remains unexplored. Most reviewed approaches either require high-frequency data or overlook real-time encrypted computation. Addressing these limitations, this study introduces a lightweight deep learning pipeline designed for 25 Hz amplitude-only CSI, enhanced by denoising filters, PCA-based compression, and CKKS-based homomorphic encryption, advancing both performance and privacy for real-world, non-intrusive healthcare monitoring. A detailed comparison of relevant existing works is provided in Table I to contextualize these contributions.

TABLE I
COMPARATIVE LITERATURE REVIEW ON CSI-BASED VITAL SIGN MONITORING AND PRIVACY-PRESERVING AI

Ref.	Study Description	Dataset & Signal Type	Techniques Used	Reported Performance	Identified Limitations
[15]	Digital twin model for respiration using CSI	Proprietary CSI at 120Hz	Signal processing, PCA, ML	Accuracy: 96.9%	High packet rate, no encryption, noise-prone
[28]	HAR using reduced-dimension CSI	HAR dataset	PCA, Decomposition, ML	Accuracy: ~94%	Degrades in multi-user setup, lacks encryption
[23]	BR and HR classification using FFT on CSI	Public CSI at 25Hz (17 subjects)	FFT, Spectral Analysis	Breathing Rate (BR): 99.18%, Heart Rate (HR): 56.55%	Low HR accuracy, no DL
[26]	HR detection using CSI phase + amplitude	CSI at 1000Hz	Hybrid ML model	Accuracy: 96.8%	Requires high-frequency hardware
[27]	PCA, ML for vital signs via CSI	Multi-subject CSI	PCA + RF/SVM	Accuracy: ~95%	Hardware dependency, lacks encryption
[29]	Visual CSI classification via spectrogram CNN	CSI visual maps	Spectrogram, CNN	Accuracy: 92–93%	High compute cost, no privacy
[30]	Respiration monitoring for elderly via WiFi CSI	CSI at 120Hz	Statistical, Amplitude model	Accuracy: 96.9%	No encryption, scalability not tested
[31]	ML-based CSI pipeline for tracking patient vitals	Simulated CSI + wearable data	Preprocessing, ML classification	Accuracy: ~90% (BR/HR)	Needs wearable calibration, no encryption
[33]	Privacy-preserving DL model for healthcare	Simulated data	HE, DL	Accuracy: 95%	Not tailored for CSI; time-series neglected

III. PRELIMINARIES

A. Channel State Information (CSI):

CSI is a fundamental concept in wireless communications, representing the channel characteristics between a transmitter (Tx) and a receiver (Rx) over specific frequencies. Wi-Fi systems that use the IEEE 802.11 standards, particularly 802.11n/ac, employ orthogonal frequency division multiplexing (OFDM) and multiple input multiple output (MIMO) technologies. These technologies improve data throughput and reliability by splitting the signal into multiple sub-carriers and transmitting it across multiple antennas [37]. In orthogonal frequency division multiplexing (OFDM), the transmission bandwidth is divided into multiple orthogonal subcarriers, each providing distinct frequency responses. For an M -receiver and N -transmitter antenna system, the CSI for a subcarrier k is represented as an $M \times N$ matrix:

$$\mathbf{H}_{i,k} = |H_{i,k}| e^{j\angle H_{i,k}} \quad (1)$$

Here, in Equation 1, $H_{i,k}$ represents the channel frequency response (CFR) for the subcarrier k corresponding to the i -th packet index between the transmitter and the receiver antenna. $|H_{i,k}|$ denotes the amplitude of the CSI response, indicating signal strength, while $\angle H_{i,k}$ represents the phase shift of the signal, reflecting time delays and propagation characteristics.

B. Homomorphic Encryption

Homomorphic encryption is a cryptographic process that enables computations to be performed directly on encrypted data without requiring decryption [38]. This ensures the privacy and security of sensitive information during processing,

making HE a cornerstone for applications where confidentiality is paramount, such as secure data sharing, healthcare analytics, and cloud computing.

HE operates through four fundamental components:

- **Key Generation (KeyGen):** This step generates a pair of keys: a public key (pk) for encryption and a secret key (sk) for decryption. The keys are used to encrypt and decrypt data securely while ensuring only authorized parties can access the original information.
- **Encryption (Encrypt):** The plaintext message m is transformed into ciphertext c using the public key:

$$c = E(m, pk) \quad (2)$$

This ciphertext can then be securely stored or transmitted without exposing the underlying plaintext.

- **Decryption (Decrypt):** The secret key is used to decrypt the result, recovering the plaintext output m_{result} :

$$m_{result} = D(C_{result}, sk) \quad (3)$$

If the scheme is implemented correctly, it m_{result} matches the output of the same function applied to the original plaintexts:

$$m_{result} = f(m_1, m_2) \quad (4)$$

1) **The CKKS Encryption Scheme:** The CKKS (Cheon-Kim-Kim-Song) scheme is a type of FHE designed for approximate arithmetic on real or complex numbers. It is particularly useful for applications involving floating-point computations, such as signal processing and encrypted analytics. CKKS supports addition and multiplication on encrypted real-valued data and enables high-precision computations with a trade-off in

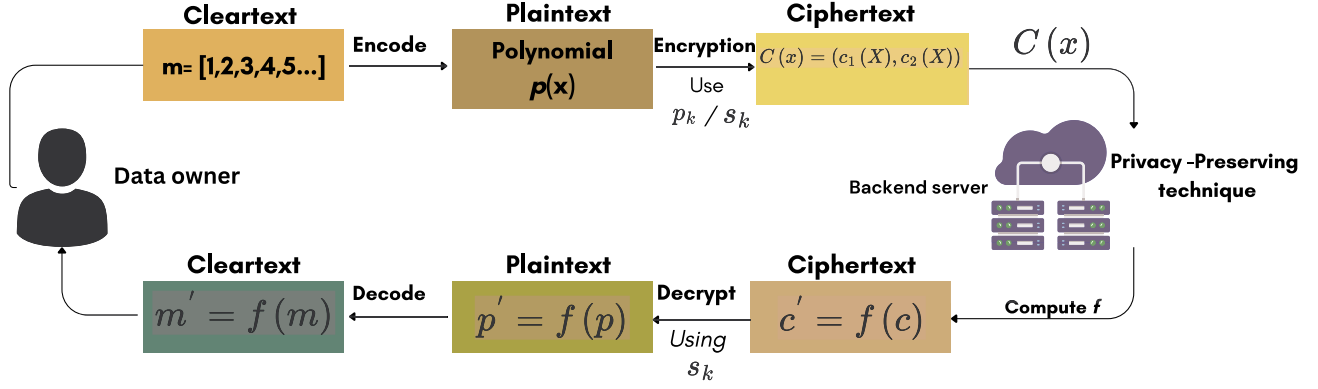


Fig. 1. Architecture for CKKS-based homomorphic encryption.

exactness due to scaling. In the CKKS scheme, a plaintext polynomial $m(X)$ is encoded into a vector of evaluations at specific points, typically roots of unity. The encoding process can be represented as:

$$\text{Enc}(m) = c = (c_0, c_1, \dots, c_{N-1}) \in \mathbb{Z}_q^N \quad (5)$$

where, $c_j = m(\zeta^j) + e_j$ for $j = 0, 1, \dots, N-1$. and ζ is a primitive N -th root of the unity. The e_j terms represent small errors introduced during the encryption process to ensure security. The decryption process can be expressed as:

$$\text{Dec}(c) = m(X) + e \quad (6)$$

where e is the accumulated error from the encryption process. The goal is to ensure that the error e remains small enough so that the significant figures of the original message m are preserved [39]. Here, Fig. 1 resembles the CKKS scheme, where data is encrypted, sent to a cloud server for computation on ciphertexts, and the encrypted results are returned for decryption by the data owner. It is also important to highlight a key limitation of the CKKS encryption scheme—noise accumulation. As encrypted operations are performed, the underlying ciphertexts accumulate noise, which limits the number of operations that can be performed before decryption becomes inaccurate. To address this, bootstrapping is required to refresh the ciphertext and reduce noise, enabling continued encrypted computation. However, this bootstrapping process is computationally expensive and further contributes to latency and resource usage. Although our current model is designed to avoid deep computations requiring bootstrapping, this limitation poses a scalability challenge for more complex models.

IV. METHODOLOGY

Our proposed framework mainly consists of two parts: one is training and classification on plain data shown in Fig. 2(a). The other one is to encrypt the data with a homomorphic encryption scheme, then use the trained model on encrypted data and perform calculations on the server side, followed by decryption and softmax approximation on the client side for classification on encrypted data as shown in Fig. 2(b). The framework has two sub-parts, namely data preprocessing and feature extraction. We used the CKKS encryption scheme via TenSEAL, as it offers efficient and accurate computation on encrypted data, making it

well-suited for privacy-preserving machine learning compared to other schemes. Researchers continue to develop more efficient privacy-preserving frameworks [34]. To reduce inference time, we employed the square activation function, which is the simplest nonlinear transformation. The CSI signal is utilized for non-intrusive physiological data collection and serves as the primary input for both feature extraction and encrypted inference in the system architecture. A detailed discussion of these parts is as follows:

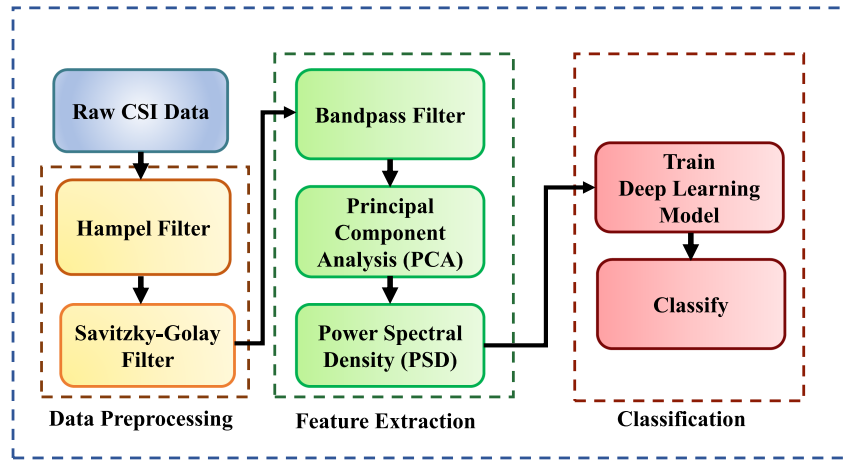
A. Data Preprocessing

Initially, CSI amplitude data is collected over a defined time window. In our study, we selected a window length of 40 seconds, as recommended in [23]. Following data collection, preprocessing is performed to remove outliers and reduce noise, ensuring that key features of the captured CSI amplitude data are accurately extracted. The data preprocessing involves two steps: Hampel filter and Savitzky-Golay filter. Accurate classification of breathing and heart rates requires reliable extraction of harmonic components from the signal. However, outliers in CSI data often hinder this process. To mitigate this, we apply the Hampel filter, an effective outlier removal technique recommended in [23] and validated on this dataset. For additional smoothing, Savitzky-Golay filtering is employed, as supported by previous studies [14], [23]. A detailed explanation of these steps is provided below.

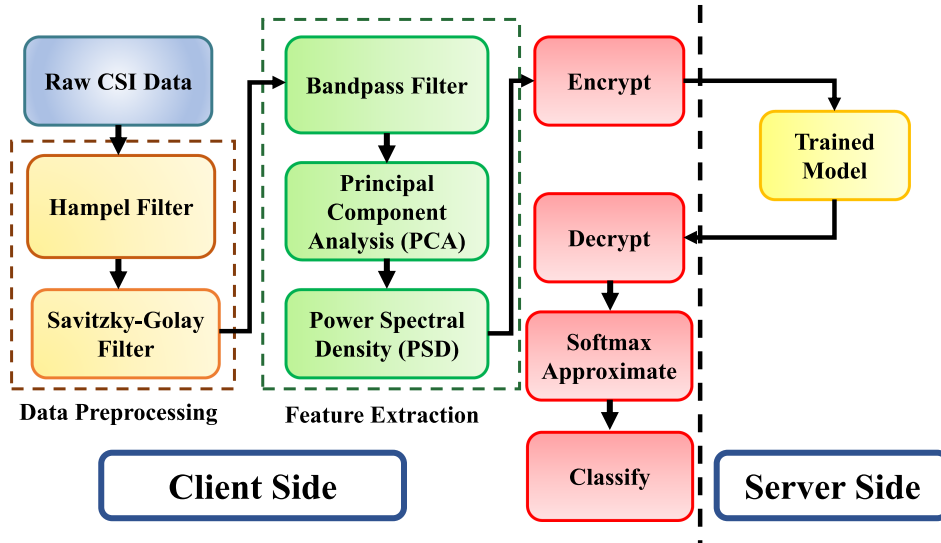
1) *Hampel Filter*: A standard Hampel filter identifies outliers for a given time window and replaces it with the median data of that window [40]. The filter operates by sliding a window of size k over the data and comparing each amplitude data point x_i of the CSI sequence with the local median m_i of the window. If the value $|x_i - m_i|$ exceeds a threshold, the value x_i is replaced by the median. The threshold is defined as n_σ times the standard deviation σ_i within the window. Specifically, for each point in the signal, the Hampel filter performs the following:

$$\tilde{x}_i = \begin{cases} m_i & \text{if } |x_i - m_i| \leq n_\sigma \cdot \sigma_i \\ x_i & \text{otherwise} \end{cases} \quad (7)$$

In our case, we set the threshold value n_σ to 2 and window length k to 200. A window of 40 seconds with a sampling rate of 25 Hz results in 1000 samples per sub-carrier. Each of these signals is



(a) Training phase with plain data.



(b) Prediction phase with encrypted data.

Fig. 2. Components of the proposed framework are (a) the training phase with plain data and (b) the prediction phase with encrypted data. In (b), a dotted vertical line separates the client-side operations (data preprocessing, feature extraction, encryption, decryption, softmax approximation) from the server-side operations (inference using the trained model).

first processed using the Hampel filter, and the output is then passed through the Savitzky-Golay filter.

2) *Savitzky-Golay Filter*: The Savitzky-Golay filter is a low-pass filter based on local least-squares polynomial approximation [41], [42]. It applies a polynomial smoothing of degree p over a moving window of size l centered around each data point. A set of coefficients derived from least-squares polynomial fitting is convoluted over the CSI amplitude data to get smooth signals. In our study, we set the polynomial degree, p to value 3 over a moving window of size 51 as recommended in [23].

B. Feature Extraction

The feature extraction process refers to extracting useful information from the given signal for accurate classification. As shown in Fig. 2(a), the feature extraction process involves

3 steps. These are the bandpass filter, PCA, and power spectral density. In the following, the steps are briefly discussed.

1) *Bandpass Filter*: An adult normally breathes at a range between 12 Breaths Per Minute (BrPM) and 20 BrPM at rest (0.2 Hz to 0.33 Hz). And the heart rate is between 60 Beats Per Minute (BPM) and 100 BPM (1 Hz to 1.67 Hz). So, in order to get the signal of interest, the cut-off frequency of 0.1 Hz and 0.4 Hz has been set for breathing rate and 1 Hz and 2 Hz for heart rate signal. Study in [15] showed that elliptic filters provide superior performance in classifying breathing rate from CSI data. For this reason, We utilized the elliptic filter for bandpass filtering of the captured amplitude data of CSI.

2) *PCA and Power Spectral Density*: Deep learning models are well-suited for capturing complex relationships within datasets. However, working with high-dimensional inputs often leads to increased computational load and may introduce

redundancy. CSI amplitude data typically falls into this category, as it includes sequences from multiple sub-carriers and antennas. To address this, PCA is applied to reduce dimensionality while retaining the essential information. PCA does this by identifying the directions in which the data varies the most and projecting the data onto these directions, ensuring that the majority of the signal's important variations are preserved and irrelevant variations are minimized. In this work, we considered 40 seconds of CSI amplitude data captured using 30 sub-carriers, 1 transmitting antenna, and 3 receiving antennas. Given a sampling frequency f_s , the resulting input shape is $(40 \times f_s, 30 \times 1 \times 3)$. With $f_s = 25$ Hz, this yields a shape of $(1000, 90)$. Using PCA, we reduced the dimensionality to $(1000, 5)$ by selecting the top 5 principal components that together accounted for over 95% of the total variance. These components form 5 time-series signals of 1000 samples each, which are then used to compute the PSD. For breathing rate estimation, we limited the PSD range to 0–40 BrPM (0–0.66 Hz). For heart rate signals, we focused on 50–90 BPM (0.833–1.5 Hz), which contained the most relevant frequency components. As a result, the final PSD input sizes were $(28, 5)$ for breathing rate and $(27, 5)$ for heart rate estimation.

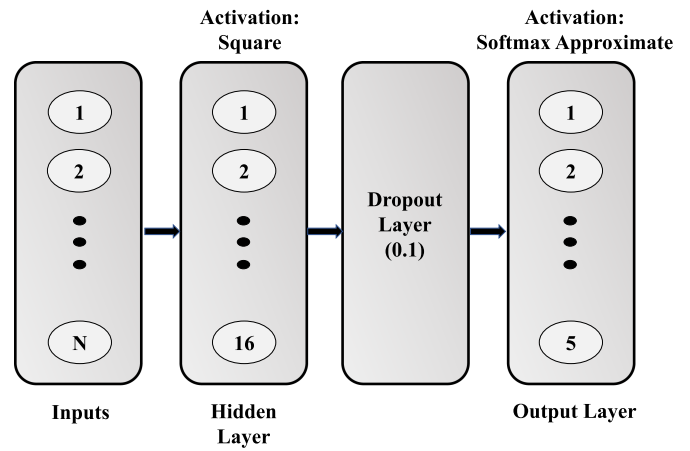
C. Homomorphic Encryption-Friendly Deep Learning Model

For classifying the breathing rate and heart rate, we utilized a simple, fully connected layer-based model that only contains one hidden layer with 16 neurons. However, the activation functions of the model need to be customized to make it homomorphic encryption friendly. Traditional activation functions such as ReLU (Rectified Linear Unit), Softmax, etc. do not support homomorphic encryption [19]. In our proposed model, we utilized the square activation in the hidden layer and the softmax approximation at the last layer for classification. The square activation function is chosen specifically because it is compatible with CKKS-based homomorphic encryption, allowing computations to be securely executed on encrypted data without any risk of data leakage on the server side. The softmax approximation can be defined by the following equations.

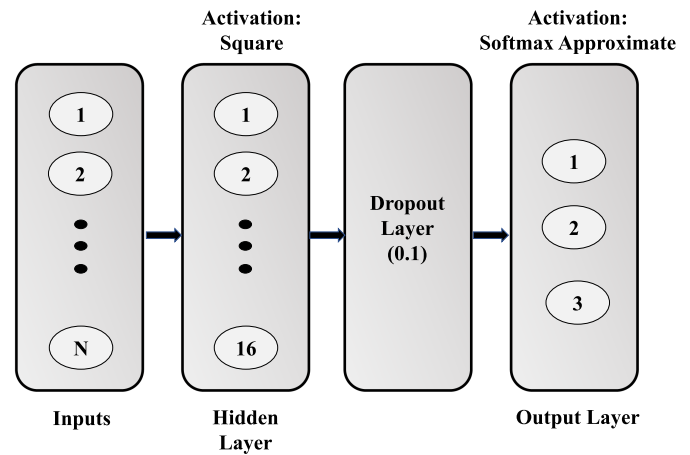
$$\exp_approx(x) = 1 + x + \frac{x^2}{2} \quad (8)$$

$$\text{softmax_approx}(x) = \frac{\exp_approx(x)}{\sum_i \exp_approx(x_i)} \quad (9)$$

Here $\exp_approx(x)$ represents the second-order polynomial approximation of $\exp(x)$, where x is the input vector. The term $\sum_i \exp_approx(x_i)$ denotes the summation of the approximated exponential values over all elements of x . Finally, $\text{softmax_approx}(x)$ is the resulting normalized approximation of the softmax function. The architecture of our proposed model for breathing rate and heart rate classification is shown in Fig. 3. Initially, we normalize the 2D input feature matrix by dividing each element by the maximum value of the entire matrix. The feature matrix is then flattened into a 1D input for feeding into the neural network. As seen in Table II and Table III, both models



(a) Model for breathing rate classification.



(b) Model for heart rate classification.

Fig. 3. Deep learning models for classifying breathing rate and heart rate.

TABLE II
SUMMARY OF THE DEEP LEARNING MODEL FOR BREATHING RATE CLASSIFICATION

Layer (Type)	Output Shape	Parameters No.
Input Layer	(None, 28, 5)	0
Flatten	(None, 140)	0
Dense	(None, 16)	2,256
Dropout	(None, 16)	0
Dense	(None, 5)	85
Total		2,341

TABLE III
SUMMARY OF THE DEEP LEARNING MODEL FOR HEART RATE CLASSIFICATION

Layer (Type)	Output Shape	Parameters No.
Input Layer	(None, 27, 5)	0
Flatten	(None, 135)	0
Dense	(None, 16)	2,176
Dropout	(None, 16)	0
Dense	(None, 3)	51
Total		2,227

TABLE IV

DETAILS OF HYPERPARAMETERS FOR TRAINING THE BREATHING RATE CLASSIFICATION MODEL

Hyperparameter	Value
Learning Rate	0.0015
Batch Size	32
Number of Epochs	120
Optimizer	Nadam
Activation Function	Square, Softmax approximate

TABLE V

DETAILS OF HYPERPARAMETERS FOR TRAINING THE HEART RATE CLASSIFICATION MODEL

Hyperparameter	Value
Learning Rate	0.00225
Batch Size	16
Number of Epochs	60
Optimizer	Nadam
Activation Function	Square, Softmax approximate

have fewer than 2400 parameters, making them lightweight and computationally efficient. This lightweight architecture, along with polynomial-compatible activation functions, ensures that encrypted inference can be performed on the server securely and efficiently using CKKS, without exposing any sensitive information. When implementing a homomorphic encryption-based neural network model, the computation time can be significantly higher [43]. Therefore, a lightweight model like ours, combined with the proposed activation functions, is a suitable choice for a homomorphic encryption-based system. We first trained our model with plain data and then used the trained model for predictions on encrypted data. The details of the model's hyperparameters are provided in Table IV and Table V.

D. Homomorphic Encryption With TenSEAL

As depicted in Fig. 2(b), the input feature data should be encrypted on the client side before being transmitted to the server for prediction. On the server side, the trained model processes the encrypted data and performs computations. The computed encrypted results are then sent back to the client, where decryption and softmax approximation are applied to classify breathing rate and heart rate. While working with encryption in a privacy-preserving framework, it is important to acknowledge that inference on encrypted data is computationally expensive and may introduce potential latency. Researchers are actively exploring optimal parameter selection by systematically varying encryption-related parameters while monitoring model performance [44]. To mitigate computational costs, model efficiency can be improved by designing a lightweight architecture and carefully selecting only the most effective features. In our study, we carefully selected the encryption parameters to ensure that the accuracy of inference on encrypted data closely matches that on plain data, while keeping the inference time per sample below 0.25 seconds. To implement this privacy-preserving encryption-based approach, we employed the CKKS homomorphic encryption scheme via the TenSEAL library. TenSEAL is an open-source framework that leverages Microsoft SEAL to implement the CKKS-based homomorphic

TABLE VI

ENCRYPTION PARAMETERS FOR CKKS SCHEME

Parameter	Value
Polynomial Modulus Degree	8192
Coefficient Modulus Sizes	{50, 35, 35, 35, 50}
Scale	2^{35}

Algorithm 1: Privacy-Preserving Inference Using Homomorphic Encryption.

Require: Raw CSI signal $\mathbf{X}$, TenSEAL context with CKKS scheme, trained model parameters $\{\mathbf{W}_i, \mathbf{b}_i\}$, public and secret keys

Ensure: Predicted class label $\hat{y}$

- 1: **Client Side (Feature Extraction & Encryption):**
- 2: Apply Hampel filter, Savitzky-Golay filter, and bandpass filter to $\mathbf{X}$
- 3: Perform PCA: $\mathbf{X}_{\text{pca}} \leftarrow \text{PCA}(\mathbf{X})$
- 4: Compute PSD features: $\mathbf{X}_{\text{psd}} \leftarrow \text{PSD}(\mathbf{X}_{\text{pca}})$
- 5: Fuse and normalize features:
 $\mathbf{m} \leftarrow \text{Normalize}([\mathbf{X}_{\text{pca}}, \mathbf{X}_{\text{psd}}])$
- 6: Encode $\mathbf{m}$ into polynomial plaintext $P(x)$ using CKKS encoding
- 7: Encrypt $P(x)$ with public key: $\mathbf{c} \leftarrow \text{Encrypt}(P(x))$
- 8: Transmit ciphertext $\mathbf{c}$ to backend server
- 9: **Backend Server (Encrypted Inference):**
- 10: Load encrypted model parameters $\{\mathbf{W}_i, \mathbf{b}_i\}$
- 11: Perform homomorphic evaluation:
- 12: **for** each layer i in model **do**
- 13: $\mathbf{c} \leftarrow \mathbf{c} \cdot \mathbf{W}_i + \mathbf{b}_i$ // HE matmul and addition
- 14: $\mathbf{c} \leftarrow \mathbf{c}^2$ // Replace ReLU with square activation
- 15: **end for**
- 16: Send encrypted output $\hat{\mathbf{c}}$ to client
- 17: **Client Side (Decryption & Prediction):**
- 18: Decrypt output: $\hat{\mathbf{y}} \leftarrow \text{Decrypt}(\hat{\mathbf{c}})$
- 19: Return predicted label: $\hat{y} \leftarrow \arg \max(\hat{\mathbf{y}})$

encryption scheme [38]. In order to implement the CKKS-based homomorphic encryption scheme, we need to define some parameters for creating a TenSEAL context. The TenSEAL context holds the key components for encryption, generating secret keys, decryption, and many more [38]. For more detailed discussion about the intuitions of these parameters, see the papers [38], [39]. In our work, we utilized the parameters in Table VI.

E. Homomorphic Encryption-Enabled Deep Learning Inference Pipeline

To ensure data confidentiality during inference, we propose a client-server pipeline where the feature vector is encrypted using CKKS homomorphic encryption before transmission. This enables approximate arithmetic on encrypted real-valued data. The server performs inference with a lightweight, HE-friendly deep learning model directly on ciphertext, without accessing plaintext. Only the client holds the decryption key, maintaining data privacy. The full pipeline—feature extraction, encryption, encrypted inference, and decryption—is detailed in Algorithm 1.

TABLE VII
BREATHING RATE CLASS LABELS

Breathing rate range	Class label
≤ 10 BrPM	1
11–13 BrPM	2
14–16 BrPM	3
17–19 BrPM	4
≥ 20 BrPM	5

TABLE VIII
HEART RATE CLASS LABELS

Heart rate range	Class label
≤ 60 BPM	1
61–80 BPM	2
≥ 80 BPM	3

V. IMPLEMENTATION AND PERFORMANCE EVALUATION

In our study, we utilized a system with a 12th Gen Intel(R) Core(TM) i5-12400F and 16 GB RAM. The software environment consisted of Python 3.10.9, TensorFlow 2.10, Tenseal 0.3.15, and CUDA 12.8 running on the Windows 10 operating system. The deep learning models were developed using Jupyter Notebooks and executed within a virtual Anaconda environment, and data preprocessing and feature extraction steps were done in MATLAB R2023a.

A. Dataset Description

For the sake of evaluation of our proposed framework, we utilized the publicly available dataset prepared by the Laboratory of Data Science and Engineering of Universidad Autónoma de Baja California [23]. Here, 17 healthy persons of different age groups participated in giving CSI data. The data collection was carried out utilizing two mini-PC Intel NUC D54250WYKH, equipped with an Intel 5300 NIC and running Ubuntu 14.04. Here, one antenna was at the transmitting side and three were on the receiving side. The packet rate of their CSI data collection was set to 25 per second.

For breathing rate classification, participants controlled their breathing rate following the beats of a metronome, which varied between 18 and 42 beats per minute (corresponding to 9 to 21 breaths per minute). Five sets of data were collected from each participant for different rhythms, each rhythm lasting 5 minutes. Ground truth breathing rates were obtained by spectral analysis of data recorded by a triaxial accelerometer placed on the participant's chest.

For heart rate classification, participants remained in the same position and wore a Polar H9 sensor to provide ground-truth heart rate measurements. A single 5-minute data set was collected for each participant in this setup. The logged outputs of the CSI Server module were used for further data analysis. The details of the breathing rate and heart rate labels are provided in Tables VII and VIII.

B. Performance Evaluation Metrics

To evaluate the classification performance of our proposed model, we utilized standard metrics, including accuracy,

precision, recall, and F1-score. Initially, data from 13 patients was used for training, while the data from the remaining 4 patients is reserved for testing and evaluation as done in [23]. Additionally, 20% of the training data have been set aside as a validation set. These metrics are defined as follows:

1) **Accuracy**: It evaluates the overall correctness of the model and is defined as the ratio of correctly predicted instances to the total number of instances.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (10)$$

Here, TP, TN, FP, and FN denote True Positives, True Negatives, False Positives, and False Negatives, respectively. TP (True Positives) are correctly predicted positive cases, TN (True Negatives) are correctly predicted negative cases, FP (False Positives) are incorrectly predicted positive cases, and FN (False Negatives) are incorrectly predicted negative cases.

2) **Precision**: It evaluates the proportion of correctly predicted positive instances out of all instances predicted as positive.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (11)$$

A high precision indicates fewer False Positives, making it important in scenarios where the cost of a false alarm is crucial.

3) **Recall**: The metric, is also known as sensitivity or True Positive Rate quantifies the model's ability to correctly identify positive instances:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (12)$$

A higher recall is crucial in applications where missing positive cases (False Negatives) has severe consequences.

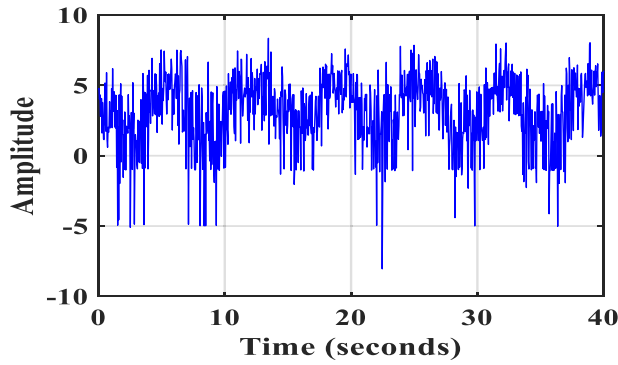
4) **F1-Score**: The parameter indicates the harmonic mean of precision and recall, showing a balanced metric when there is an uneven class distribution.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

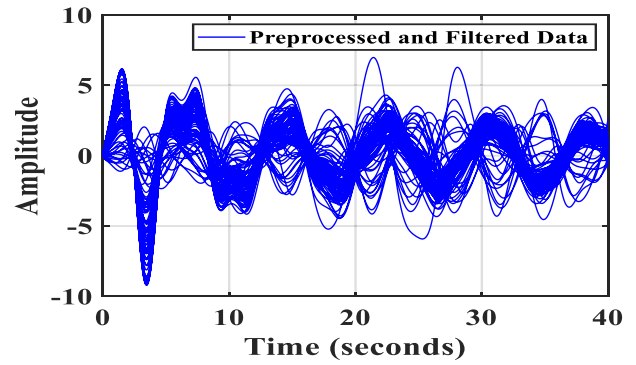
Overall, these metrics give a comprehensive overview of how well the classification model performs. Accuracy provides a broad view of performance, while Precision and Recall focus on different aspects of it, and the F1-Score strikes a balance between them. Depending on the problem, False Positives and False Negatives would be viewed differently, and thus the metric must be selected accordingly.

C. Breathing Rate Classification Performance

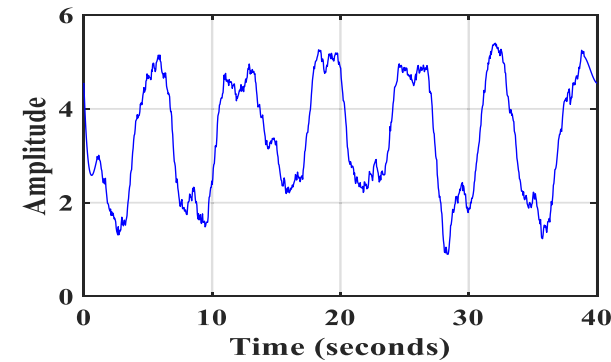
1) **Data Preprocessing and Filtering**: In Fig. 4, the results from preprocessing and filtering steps are depicted for a single CSI signal of 40 seconds. The raw CSI data in Fig. 4(a) clearly has lots of random environmental noises and outliers, which would potentially cause issues while extracting desired features. After the Hampel and SG filtering, we get a signal where most of the outliers and unwanted noise get removed, as shown in Fig. 4(b). Then the smoothed data is passed through the bandpass filter as discussed in Section IV-B1. There we get smooth and filtered data most suitable for feature extraction.



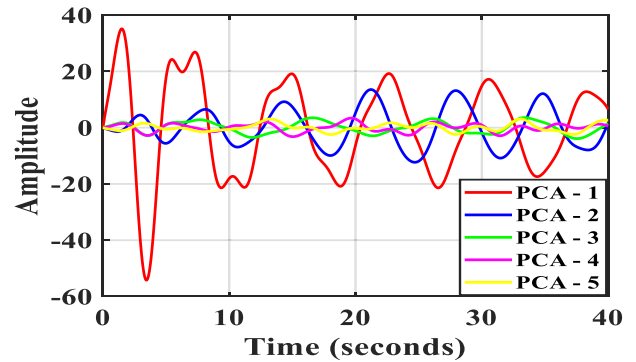
(a) Breathing rate - Raw CSI.



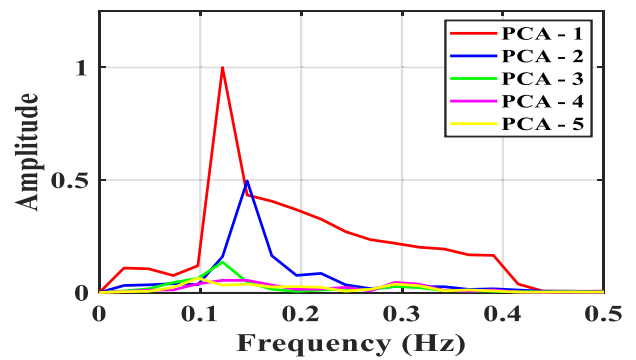
(b) Breathing rate - After Hampel and SG filter.



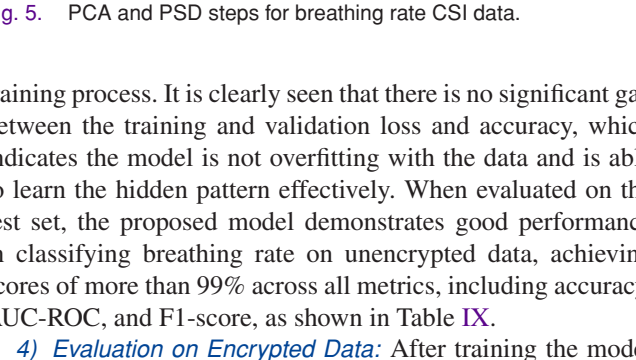
(c) Breathing rate - After Bandpass filter.



(a) Filtered breathing rate signal before PCA.



(b) Filtered breathing rate signal after PCA.



(c) PSD of PCA for breathing rate signals.

Fig. 4. Data preprocessing and filtering steps for breathing rate CSI.

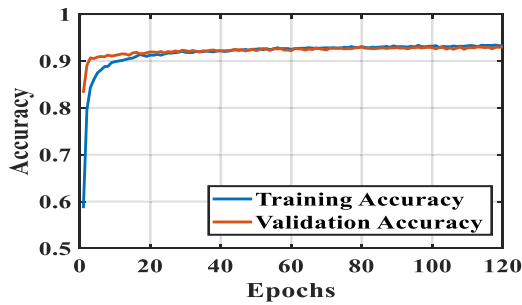
Fig. 5. PCA and PSD steps for breathing rate CSI data.

2) PCA and PSD Feature Extraction: The experimental dataset consists of CSI amplitude data of 30 subcarriers with 1 transmitting antennas and 3 receiving antennas. Therefore, we get $30 \times 1 \times 3 = 90$ streams of CSI signal as shown in Fig. 5(a). After performing the preprocessing and filtering, we utilized PCA for reducing the number of datastreams from 90 to 5 as depicted in Fig. 5(b). Furthermore, we utilized the PSD of those PCA outputs as discussed in Section IV-B2. Fig. 5(c) shows the PSD of the first 5 PCA results.

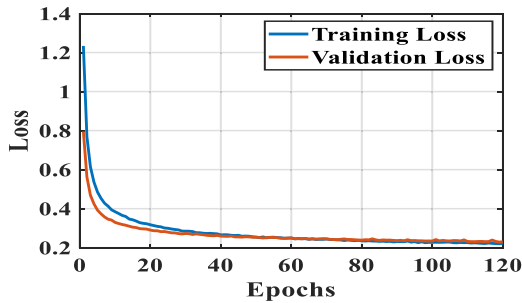
3) Training and Evaluation on Plain Data: After performing the feature extraction part, the homomorphic encryption-friendly deep learning model has been utilized for training on plain data. The model architecture and hyperparameter details are discussed in the earlier sections. In Fig. 6, we can see the

training process. It is clearly seen that there is no significant gap between the training and validation loss and accuracy, which indicates the model is not overfitting with the data and is able to learn the hidden pattern effectively. When evaluated on the test set, the proposed model demonstrates good performance in classifying breathing rate on unencrypted data, achieving scores of more than 99% across all metrics, including accuracy, AUC-ROC, and F1-score, as shown in Table IX.

4) Evaluation on Encrypted Data: After training the model on the training set, we evaluate its performance on encrypted data. First, the features of the test set are extracted, and then the test set data is encrypted. The trained model is used for both computation and prediction on this encrypted data. We implement the CKKS homomorphic encryption scheme using the TenSEAL



(a) Accuracy plot for breathing rate classification.



(b) Loss plot for breathing rate classification.

Fig. 6. Training and validation accuracy and loss for breathing rate classification.

TABLE IX

PERFORMANCE METRICS FOR BREATHING RATE CLASSIFICATION

Metric	Value (%)
Accuracy	99.4615
AUC-ROC	99.9638
F1 Score	99.4622

TABLE X

COMPARISON OF ACCURACY AND COMPUTATION TIME PER SINGLE PREDICTION FOR PLAIN AND ENCRYPTED DATA IN BREATHING RATE CLASSIFICATION

Data Type	Accuracy (%)	Time (s)
Plain	99.46	0.034 ± 0.003
Encrypted	99.44	0.232 ± 0.011

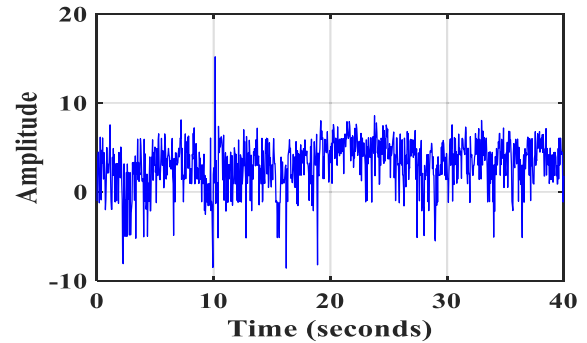
library. Since the computations are performed on encrypted data, it is expected that the model requires more computation time and may experience a slight decrease in accuracy compared to plain data. This tradeoff is inherent to encryption-based deep learning due to the approximations involved in processing encrypted data. Table X shows that the accuracy for plain and encrypted data is comparable, with values of 99.46% and 99.44%, respectively. However, the computation time per single prediction increases significantly, being approximately 7 times higher for encrypted data (0.232 ± 0.011 seconds) compared to plain data (0.034 ± 0.003 seconds).

5) Performance Comparison: In Table XI, a performance comparison is shown between our proposed framework (with and without encryption) and other studies. It is seen that our proposed model shows commendable results compared to other existing studies even with the encrypted data. The packet rate of our dataset is low compared to some studies. However, our

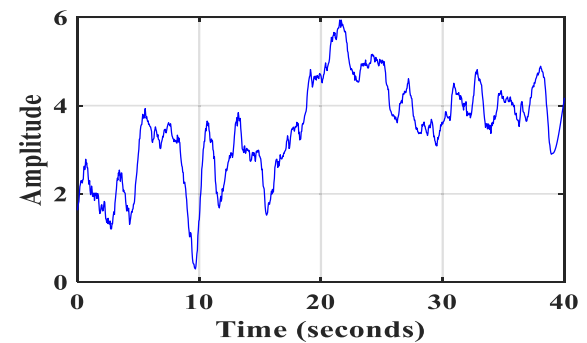
TABLE XI

PERFORMANCE COMPARISON BETWEEN OUR MODEL AND OTHER STUDIES

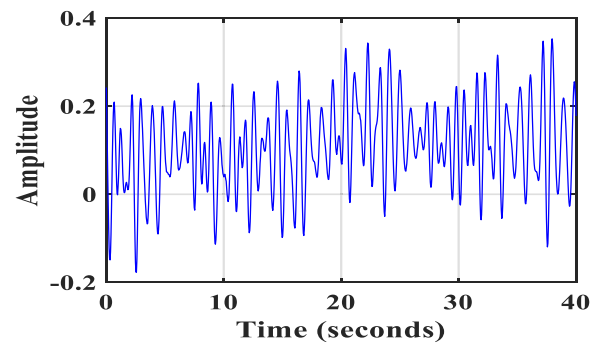
Ref.	Pkt. Rate	CSI Type	Acc (%)
[30]	120 Hz	Amplitude	96.90
[15]	120 Hz	Amplitude	96.90
[23]	25 Hz	Amplitude	99.18
Ours (Plain)	25 Hz	Amplitude	99.46
Ours (Enc.)	25 Hz	Amplitude	99.44



(a) Heart rate - Raw CSI.



(b) Heart rate - After Hampel and SG filter.



(c) Heart rate - After Bandpass filter.

Fig. 7. Data preprocessing and filtering steps for heart rate CSI.

proposed framework is still able to show good performance on that. It confirms the robustness and reliability of our proposed model, which also ensures the privacy of the data.

D. Heart Rate Classification Performance

1) Data Preprocessing and Filtering: In Fig. 7, the results from preprocessing and filtering steps are depicted for a single CSI signal of 40 seconds, the same as the breathing rate. The raw

TABLE XII
PERFORMANCE METRICS FOR HEART RATE CLASSIFICATION

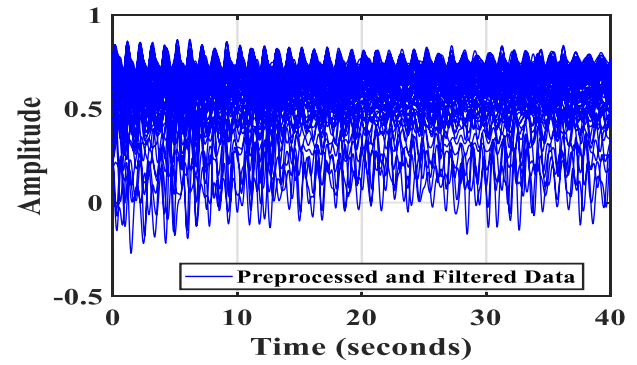
Metric	Value (%)
Accuracy	70.8654
AUC-ROC	67.7328
F1 Score	70.7704

CSI data in Fig. 7(a) clearly has lots of random environmental noises and outliers like the breathing rate signal. After the Hampel and SG filtering, we get a signal where most of the outliers and unwanted noise get removed, as shown in Fig. 7(b). Then the smoothed data is passed through the bandpass filter as discussed in Section IV-B1. There we get smooth and filtered data most suitable for feature extraction. It is worth noticing that the cleaned signal in Fig. 7(c) does not show a clear periodic pattern as we have seen in breathing rate smooth signals in Fig. 4(c).

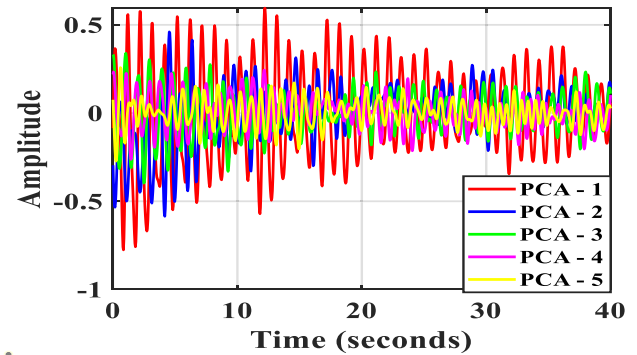
2) *PCA and PSD Feature Extraction*: The experimental heart rate CSI dataset also consists of CSI amplitude data of 30 subcarriers with 1 transmitting antennas and 3 receiving antennas. Consequently, we obtain $30 \times 1 \times 3 = 90$ streams of CSI signals, as shown in Fig. 8(a). Following preprocessing and filtering, PCA was applied to reduce the number of data streams from 90 to 5, as depicted in Fig. 8(b). Furthermore, the PSD of these PCA outputs was computed, as discussed in Section IV-B2. Fig. 8(c) shows the PSD of the first 5 PCA outputs. While this approach is similar to what was applied for the breathing rate, other than the cutoff frequencies of the bandpass filter. The analysis of the results reveals a significant difference. Unlike the breathing rate, where the PSD of the PCA outputs exhibits a dominant peak corresponding to the desired frequency, the PSD for heart rate shows multiple peaks across different frequency regions. This indicates potential issues in acquiring a clean heart rate signal from the dataset.

3) *Training and Evaluation on Plain Data*: Similarly to breathing rate, the PSD feature of PCA has been utilized in the homomorphic-friendly deep learning model to classify heart rate into three different classes. For heart rate, some model hyperparameters differ, with the epoch count reduced to 60. This adjustment is due to the heart rate signal often being contaminated by noisy, unwanted signals, which diminishes the effectiveness of the feature extraction process. As a result, the model risks overfitting and struggles to generalize to unknown data. While Fig. 9 shows a low performance gap between the training and validation sets, the model demonstrates weaker performance on new data when trained for more epochs. Table XII shows that the model can get around 70% accuracy on the test set, which is not as high as we got for the breathing rate classification model.

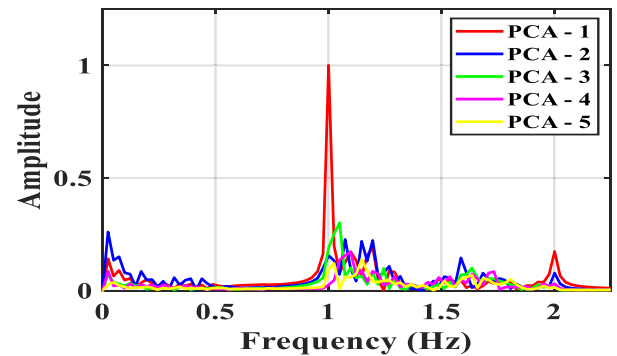
4) *Evaluation on Encrypted Data*: After training the model on the training set, its performance is evaluated on encrypted data using the CKKS homomorphic encryption scheme via TenSEAL, where, similar to the breathing rate classification, increased computation time and a slight accuracy reduction are expected compared to plain data. Table XIII demonstrates that the accuracy for both plain and encrypted data is quite similar, with values of 70.87% for plain data and 70.77% for encrypted



(a) Filtered Heart rate signal before PCA.



(b) Filtered Heart rate signal after PCA.



(c) PSD of PCA for Heart rate signals

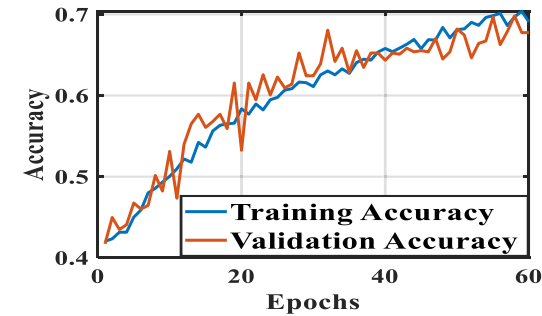
Fig. 8. PCA and PSD steps for heart rate CSI data.

TABLE XIII
COMPARISON OF ACCURACY AND COMPUTATION TIME PER SINGLE PREDICTION FOR PLAIN AND ENCRYPTED DATA IN HEART RATE CLASSIFICATION

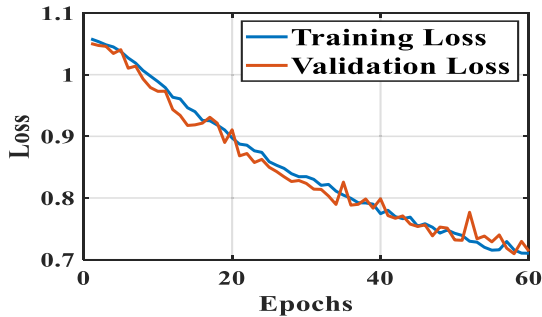
Data Type	Accuracy (%)	Time (s)
Plain	70.87	0.034 ± 0.003
Encrypted	70.77	0.224 ± 0.007

data. However, similar to previous results, the computation time per prediction increases considerably, with encrypted data taking roughly 7 times longer (0.224 ± 0.007 seconds) than plain data (0.034 ± 0.003 seconds).

5) *Performance Comparison*: Table XIV compares heart rate detection performance across different studies, including



(a) Accuracy plot for heart rate classification.



(b) Loss plot for heart rate classification.

Fig. 9. Training and validation accuracy and loss for heart rate classification.

TABLE XIV

PERFORMANCE COMPARISON BETWEEN OUR MODEL AND OTHER STUDIES

Ref.	Pkt. Rate	CSI Type	Acc (%)
[26]	1000 Hz	Amplitude, Phase	96.80
[45]	500 Hz	Amplitude	94.22
[23]	25 Hz	Amplitude	56.55
Ours (Plain)	25 Hz	Amplitude	70.87
Ours (Enc.)	25 Hz	Amplitude	70.77

ours. While some studies report accuracies above 90%, our model achieves around 70%. A closer look at Table XIV shows that these systems use packet rates of 500 Hz or higher. Additionally, one study improves accuracy by using CSI phase data [26]. In contrast, our study uses data with a packet rate of 25 packets per second, which is about 1/20 the packet rate used in other studies. A similar study with a low packet rate achieved around 56% accuracy, which our framework outperforms with the same dataset. The authors of [23] also discuss these limitations. Thus, we can reasonably conclude that our framework would perform better with a higher packet rate, improving the detection of subtle movements like heart rate from the skin.

E. Sensitivity Analysis

The polynomial modulus degree in CKKS controls ciphertext size and precision; higher values increase security but also raise computation time in TenSEAL. Table XV shows how inference time per sample grows with the modulus degree, while detection accuracy remains unchanged at 99.44% for breathing rate and 70.77% for heart rate.

TABLE XV

INFERENCE TIME PER SAMPLE FOR DIFFERENT POLYNOMIAL MODULUS DEGREES

Modulus Degree	Breathing Rate (s)	Heart Rate (s)
8192	0.232 ± 0.011	0.224 ± 0.007
16384	0.5181 ± 0.014	0.5100 ± 0.017
32768	1.234 ± 0.031	1.181 ± 0.030

F. Trade-Off Between Computation, Accuracy, and Encryption in Healthcare

In privacy-sensitive healthcare environments, securing patient data during model inference is essential. Using the CKKS encryption scheme through TenSEAL, we observed a notable increase in inference time when processing encrypted data. For example, breathing rate predictions on encrypted data took around 0.232 ± 0.011 seconds per sample, compared to 0.034 ± 0.003 seconds on unencrypted data. Despite the added latency, model accuracy remained nearly unchanged—99.44% for encrypted vs. 99.46% for plain data in breathing rate detection, and 70.77% vs. 70.87% for heart rate. These findings illustrate the computational cost of encryption, even though it preserves predictive performance, posing practical considerations for deployment in time-sensitive healthcare applications.

G. System and Threat Model

Our client-server framework performs data preprocessing and feature extraction on the client side, while inference runs on the server (Fig. 3). Training uses filtered and PCA-reduced CSI data to build a plaintext model deployed for encrypted inference.

At prediction, extracted features are encrypted via CKKS and sent to the server, which computes without decrypting. The client then decrypts the results.

Threat Model: An honest-but-curious server follows protocols but may attempt data inference. CKKS encryption safeguards all client data during inference, ensuring privacy.

H. Communication Overhead

In this work, the communication overhead becomes relevant during the prediction stage, where encrypted CSI-based feature representations are sent from the client device to the server for inference. Because we employ CKKS homomorphic encryption, which expands the size of data to ensure secure computation, each encrypted feature vector becomes significantly larger than its plaintext version. Depending on the encryption configuration—such as the polynomial modulus degree—the size may vary from hundreds of kilobytes to several megabytes. Although this leads to higher data transmission requirements, it is a reasonable trade-off when prioritizing privacy in applications involving sensitive physiological information derived from CSI signals.

I. Clinical Trial Considerations

This study classifies vital signs using CSI from public datasets but lacks clinical validation. No trials have assessed its performance in real-world healthcare settings with factors like

patient movement and noise. Future work will conduct clinical trials to ensure reliability, regulatory compliance, and real-world applicability.

VI. CONCLUSION

In this paper, we explore an encryption-based, privacy-preserving deep learning framework, VitalCrypt, for tracking vital signs from CSI amplitude data. First, we extract features from the CSI data, which are then classified using our homomorphic encryption-friendly neural network model with custom activation functions designed for encrypted computation. To implement the encryption-based system, we utilize the TenSEAL library to execute CKKS homomorphic encryption and perform predictions on encrypted data. We evaluate our proposed framework using a publicly available dataset. The deep learning model is initially trained on plain data, and the trained model is then used to assess prediction performance on both encrypted and non-encrypted data. Our model classifies breathing rate and heart rate, achieving an accuracy of over 99% for breathing rate classification in both plain and encrypted settings. While encrypted inference results in a slight accuracy drop and increased computational time, it still performs commendably. However, heart rate classification shows lower accuracy, possibly due to limitations such as packet rate constraints and the absence of phase data, which we plan to address in future work. We initially trained our model using plain data to minimize computational complexity. Future work will explore training with encrypted data and evaluate the model's performance under these conditions. A significant advantage of our study is that VitalCrypt ensures data privacy while maintaining competitive classification performance. With the exponential growth of healthcare data, privacy concerns, especially for patient medical records, are becoming increasingly important. Our framework presents a promising solution for privacy-preserving patient vital sign monitoring.

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